

SPPL-200: Fluctuation Allowance (Monthly FA only) Get API technical workflow

Jira Ticket: [SPPL-200: BE | Create technical workflow for Fluctuation Allowance Get \(Monthly FA only\) API on parameter inputs page including DB details](#) DONE

- i** This page provides comprehensive information about the GET API for Monthly FA on the Input params page, to view Monthly FA details for the scenario.

The screenshot shows the 'cube' system interface for 'Global Planning System'. The 'Supply Planning' tab is active. The page title is 'AP/TMMK Line 3/Aug 2025 V1'. The 'Views' sidebar on the left lists various input categories, with 'Fluctuation Allowance' selected. The main content area is titled 'Fluctuation Allowance' and has a status of 'WORK IN PROGRESS'. It contains two sections: 'Monthly F/A' and 'Daily F/A'. The 'Monthly F/A' section has a toggle 'Apply to all months' and a value of 32%. The 'Daily F/A' section has a toggle 'Apply to all days' and a value of 20%. Both sections have a 'See Master Data' button. At the bottom, there are 'Cancel', 'Save', and 'Ready' buttons.

This screenshot shows the same 'Fluctuation Allowance' configuration page, but with more detailed settings visible. The 'Monthly F/A' section is expanded, showing a table for 'Edit Monthly Values (1-39 months)'. The table has columns for 'Month' and 'Value %'. The values for months 24, 25, 26, and 27 are 12, 10, 11, and 21 respectively. The 'Daily F/A' section is also expanded, showing a table for 'Edit Daily Allowances'. The table has columns for 'Day' and 'Value %'. The values for days 1 through 5 are all 3. The 'Status' is still 'WORK IN PROGRESS'. The 'Cancel', 'Save', and 'Ready' buttons are at the bottom.

Supply Planning GET Fluctuation Allowance API Details:

1. **Hosted URL:** <https://api.spln.cube<env>.toyota.com/<env>>
2. **API Name:** <HOSTED_URL>/v/1/1/sp/monthly-fluctuation-allowance
3. **Path Param:** N/A
4. **Query Param:** ?scenarioId=uniqueScenarioId&userEmail=userEmail

5. **Method:** GET

6. **Authorization:** Bearer token (**required**)

7. **Lambda Name:** spln-<env>-view-monthly-fluctuation-allowance-v1-1

8. **API Request Body:** N/A

9. **API Response:**

- **Success response with status 200:**

```
1  {
2    "scenarioId": "uniqueScenarioId",
3    "data": [
4      {
5        "groupId": "0c1b2a3d-1111-2222-3333-444455556666",
6        "groupName": "Group 1",
7        "vanningCenter": "TMH",
8        "fluctuationAllowance": {
9          "isUniformAcrossMonths": true,
10         "ranges": [
11           { "month": 3, "fa": 14 },
12           { "month": 4, "fa": 14 }
13         ]
14       }
15     },
16     {
17       "groupId": "0d1b2a3d-1111-2222-3333-444455557777",
18       "groupName": "Group 2",
19       "vanningCenter": "TMK",
20       "fluctuationAllowance": {
21         "isUniformAcrossMonths": false,
22         "ranges": [
23           { "month": 3, "fa": 14 },
24           { "month": 4, "fa": 10 }
25         ]
26       }
27     }
28   ],
29   "scenarioSteps": {
30     "Line Level Input": {
31       "Model Change Dates": "Completed",
32       "NAMC Allocation Plan": "Completed",
33       "NAMC Production Calendar": "Completed"
34     },
35     "Vanning Center Inputs": {
36       "Shipping Pattern": "Completed",
37       "Vanning Lead Time": "Completed",
38       "TMC Working Day Calendar": "Completed"
39     },
40     "Grouping Settings": {
41       "Grouping": "Completed",
42       "Min Max Stock": "In Progress",
43       "Fluctuation Allowance": "Not Started"
44     },
45     "other": [
46       "Simulation",
47       "Review",
48       "Reports"
49     ]
50   }
51 }
```

- **Error Response:**

- **any internal server error occurred with status 500:**

```
1 {  
2   "errorMessage": "Internal Server Error"  
3 }
```

- **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```
1 {  
2   "message": "Unauthorized"  
3 }
```

- **If request is in-valid with status 400:**

```
1 {  
2   "errorMessage": [<"ValidationError: validation error message">]  
3 }
```

- **List of possible validation error message:**

1. "ValidationError: scenarioId is required and must be a uuid."
2. "ValidationError: Scenario doesn't exist."
3. "ValidationError: userEmail is required and must be a string."
4. "ValidationError: Invalid userEmail."

10. Table:

- a. **scenarios**
- b. **grouping**
- c. **group_scenario_mapper**
- d. **monthly_fa**
- e. **scenario_step_status**

SP Fetch Fluctuation Allowance Data API Workflow:

- **Technical workflow:**

- Description:
 - i. The Supply Planning user logs in to the application.
 - ii. After the user provides creates a scenario, opens it and navigates to Fluctuation Allowance, a request is sent to the SP Get Fluctuation Allowance endpoint:
sp/monthly-fluctuation-allowance with parameters: type=monthly and scenarioId=uniqueScenarioId&userEmail=userEmail
 - iii. After successful authentication, the ***spln-view-monthly-fluctuation-allowance-v1-1*** lambda is triggered by the SP view scenario Fluctuation Allowance endpoint: ***sp/fluctuation-allowance***. If authentication fails, an error with status 401 (*refer:*

error response-**any unauthorized error occurred with status 401**) will be returned and sent to client.

- iv. After user authentication has been confirmed, the **spln-view-monthly-fluctuation-allowance-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (refer: error response: **Section if request is in-valid with status 400 and List of possible validation error message**). Below query is used to check if provided scenario is valid:

```
1 select * from supply_planning.scenarios where scenario_id = :scenarioId and
  is_active = true;
```

- v. After the request payload is successfully validated, the **spln-view-monthly-fluctuation-allowance-v1-1** lambda execute below query to get scenario Grouping & monthly FA data from the **grouping, group_scenario_mapper & monthly_fa** tables:

```
1 SELECT
2     gp.group_id as "groupId",
3     sgm.group_name AS "groupName",
4     gp.vanning_center AS "vanningCenter",
5     mfa.fa_month AS "month",
6     mfa.year AS "year"
7     mfa.apply_to_all_months as "isUniformAcrossMonths"
8 FROM supply_planning.group_scenario_mapper sgm
9 JOIN supply_planning.grouping gp
10    ON gp.group_id = sgm.group_id
11 JOIN supply_planning.monthly_fa mfa
12    ON mfa.group_id = gp.group_id
13 WHERE sgm.scenario_id = $1::uuid
14    AND sgm.is_active = TRUE
15    AND gp.is_active = TRUE;
```

In case of no monthly fa data for any active group, prepare initial data with '0' for all the scenario timeframe months & years in below response format:

```
1 {
2     "scenarioId": "uniqueScenarioId",
3     "data": [
4         {
5             "groupId": "0c1b2a3d-1111-2222-3333-444455556666",
6             "groupName": "Group 1",
7             "vanningCenter": "TMH",
8             "fluctuationAllowance": {
9                 "isUniformAcrossMonths": true,
10                 "ranges": [
11                     { "month": 3, "fa": 0 },
12                     { "month": 4, "fa": 0 }
13                 ]
14             }
15         },
16         {
17             "groupId": "0d1b2a3d-1111-2222-3333-444455557777",
18             "groupName": "Group 2",
19             "vanningCenter": "TMK",
20             "fluctuationAllowance": {
21                 "isUniformAcrossMonths": true,
```

```

22     "ranges": [
23         { "month": 3, "fa": 0 },
24         { "month": 4, "fa": 0 }
25     ]
26     }
27 }
28 ],
29 "scenarioSteps": {
30     "Line Level Input": {
31         "Model Change Dates": "Completed",
32         "NAMC Allocation Plan": "Completed",
33         "NAMC Production Calendar": "Completed"
34     },
35     "Vanning Center Inputs": {
36         "Shipping Pattern": "Completed",
37         "Vanning Lead Time": "Completed",
38         "TMC Working Day Calendar": "Completed"
39     },
40     "Grouping Settings": {
41         "Grouping": "Completed",
42         "Min Max Stock": "In Progress",
43         "Fluctuation Allowance": "Not Started"
44     },
45     "other": [
46         "Simulation",
47         "Review",
48         "Reports"
49     ]
50 }
51 }

```

vi. Below query is used to fetch view scenario menu data: ref ([SPPL-162: View Scenario Menu technical workflow document](#))

```

1  select * from supply_planning.scenario_step_status where scenario_id =
   :scenarioId;

```

vii. The retrieved data is formatted as shown below before returning the response to the client with a status code of 200:

1. If the Monthly FA data for scenario is available:

```

1  {
2    "scenarioId": "uniqueScenarioId",
3    "data": [
4      {
5        "groupId": "0c1b2a3d-1111-2222-3333-444455556666",
6        "groupName": "Group 1",
7        "vanningCenter": "TMH",
8        "fluctuationAllowance": {
9          "isUniformAcrossMonths": true,
10         "ranges": [
11           { "month": 3, "fa": 14 },
12           { "month": 4, "fa": 14 }
13         ]
14       }
15     },
16     {
17       "groupId": "0d1b2a3d-1111-2222-3333-444455557777",
18       "groupName": "Group 2",
19       "vanningCenter": "TMK",
20       "fluctuationAllowance": {
21         "isUniformAcrossMonths": false,
22         "ranges": [

```

```

23         { "month": 3, "fa": 14 },
24         { "month": 4, "fa": 10 }
25     ]
26 }
27 }
28 ],
29 "scenarioSteps": {
30     "Line Level Input": {
31         "Model Change Dates": "Completed",
32         "NAMC Allocation Plan": "Completed",
33         "NAMC Production Calendar": "Completed"
34     },
35     "Vanning Center Inputs": {
36         "Shipping Pattern": "Completed",
37         "Vanning Lead Time": "Completed",
38         "TMC Working Day Calendar": "Completed"
39     },
40     "Grouping Settings": {
41         "Grouping": "Completed",
42         "Min Max Stock": "In Progress",
43         "Fluctuation Allowance": "Not Started"
44     },
45     "other": [
46         "Simulation",
47         "Review",
48         "Reports"
49     ]
50 }
51 }

```

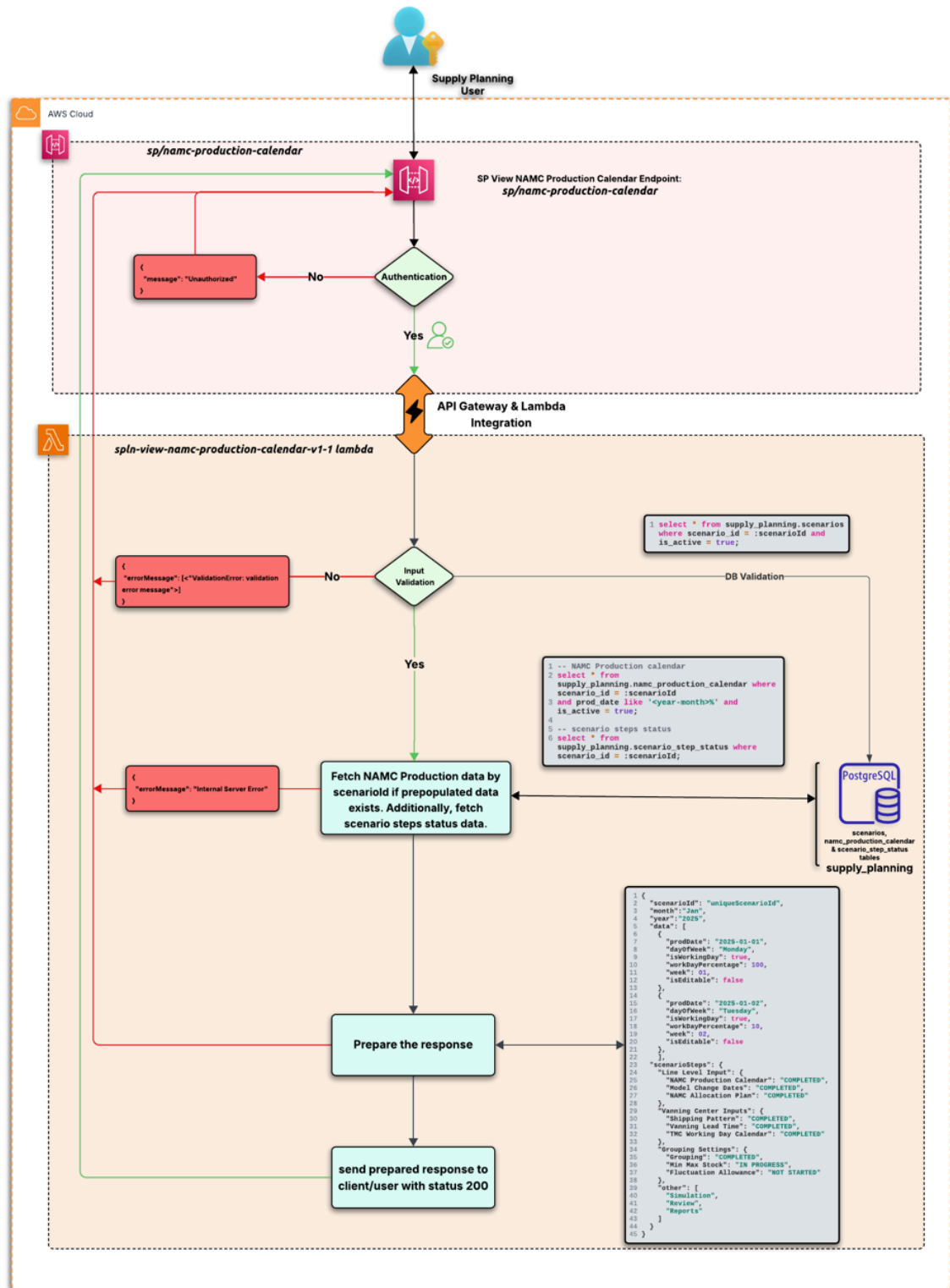
2. If Monthly FA data is not available:

```

1  {
2      "scenarioId": "uniqueScenarioId",
3      "data": [],
4      "scenarioSteps": {
5          "Line Level Input": {
6              "Model Change Dates": "COMPLETED",
7              "NAMC Allocation Plan": "COMPLETED",
8              "NAMC Production Calendar": "COMPLETED"
9          },
10         "Vanning Center Inputs": {
11             "Shipping Pattern": "COMPLETED",
12             "Vanning Lead Time": "COMPLETED",
13             "TMC Working Day Calendar": "COMPLETED"
14         },
15         "Grouping Settings": {
16             "Grouping": "COMPLETED",
17             "Min Max Stock": "IN PROGRESS",
18             "Fluctuation Allowance": "NOT STARTED"
19         },
20         "other": [
21             "Simulation",
22             "Review",
23             "Reports"
24         ]
25     }
26 }

```

- viii. If there is an error while fetching the data, an “Internal Server Error” with a status code of 500(refer: error response-**any internal server error occurred with status 500**) will be returned. Otherwise, a successful response with a status code of 200 will be sent.



SP NAMC Production Calendar Get API Technical workflow

Note:

We will be utilizing this API for both to fetch file data and previous rundown data.

DB constraints:

```
CREATE INDEX IF NOT EXISTS idx_group_scenario_mapper_scenario_active  
ON supply_planning.group_scenario_mapper (scenario_id, is_active, group_id);
```

```
CREATE INDEX IF NOT EXISTS idx_monthly_fa_scenario_group_year_month  
ON supply_planning.monthly_fa(scenario_id, group_id, year, month);
```

Queries: